

VECTOR RAG · THE BLIND SPOT

Flat retrieval is blind to the relations between entities.

Chunks in, vectors out, nearest neighbours back.
Isolated fragments, no structure, no memory of how facts connect.

BENCHMARKS

2 HotpotQA & MuSiQue
multi-hop reasoning suites

Both demand linking facts across entities.
Flat RAG fails.

$\text{cosine}(\text{query}, \text{chunk}) \cdot \text{context lost.}$

GRAPHRAG

Knowledge as a **graph** of entities and relations, not a list of chunks.

We build the graph automatically from the source documents, then traverse it to assemble a structured, multi-level context before generation.

01

Extract

LLM triple extraction

02

Build

RDF + NetworkX graph

03

Cluster

Leiden communities

04

Query

Multi-hop + LLM answer

End-to-end pipeline · extraction · RDF graph · community detection · multi-hop retrieval + LLM.

PIPELINE

Building the graph.

01 · EXTRACT

LLM triple extraction

The model reads each chunk and emits (subject, relation, object) triples, e.g. Turing · COLLABORATED_WITH · Welchman. Pronouns, duplicates and empty fields are filtered.

02 · BUILD

RDF graph · NetworkX

Triples aggregate into a directed graph. Nodes are entities, edges are typed relations in UPPER_SNAKE_CASE.

03 · CLUSTER

Leiden community detection

The graph is partitioned into thematically coherent communities via `leidenalg`, with a fallback to connected components when the library is unavailable.

MULTI-HOP TRAVERSAL

From a question to a sub-graph.

- ◆ Anchor. Match question entities to nodes. These become the BFS entry points.
- ◆ Traverse. Walk 2 hops in and out. Capped at 150 nodes for the LLM context.
- ◆ Generate. Serialise the sub-graph, hand it to the LLM. Every answer ships with a trace.

TRACE · SAMPLE

question: who decoded Enigma with Turing?

anchor → [Turing]

hop 1 → Welchman, Knox, Hut 8

hop 2 → Bletchley_Park, GC&CS

context:

Turing · COLLABORATED_WITH · Welchman

Turing · WORKED_AT · Hut 8

Welchman · MEMBER_OF · Bletchley_Park

answer → Gordon Welchman.

THE ONE MISSING PIECE

Swap text matching for semantic anchoring.

POC · TODAY

Text matching anchors

If the question doesn't mention an entity name verbatim, we find nothing. Brittle by design, surface form only.

PRODUCTION · NEXT

Embedding-based anchors

Every node carries an embedding vector. We find the nearest node to the question by similarity. Synonyms and paraphrases all welcome.

From that entry node, the rest is identical: BFS hop by hop across the same graph. The traversal doesn't change, only the door does. That single component is everything standing between this POC and a production system.